

COMPSCI5079 (M)

Cryptography & Secure Development

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Lecture 3:

Symmetric-key Cryptography Block Ciphers





Lecture 2:

Review



Main Points to take home

1. Types of attack:

- Ciphertext only
- Known (part of) plaintext and ciphertext, find the key
- Chosen plaintext

2. Attack Methods:

- Brute Force
- Letter Frequency
 - Diagram and Trigram



Main Points to take home

3. Transposition ciphers:

- Split characters into blocks of fixed length d .
- Rearrange the characters inside a block according to a key-dependent permutation.
- Decryption uses the inverse permutation to recover the plaintext.

4. Substitution ciphers:

- Provide a matching between letters in the plaintext alphabet to other letters in the same or a different alphabet for the ciphertext.
- Shift Substitution
- Multiplicative Substitution



Main Points to take home

5. Homophonic ciphers:

- A homophonic cipher has more letters in the ciphertext alphabet than in the plaintext alphabet.
 - Each plaintext letter can be replaced by a choice of several ciphertext letters (chosen at random).
 - This reduces the effect of statistics.
- A second-order homophonic encrypts two different messages, each with their own key, into one piece of ciphertext.
 - An innocent message is protected by a distress key.



Main Points to take home

5. Polyalphabet ciphers:

- A polyalphabet cipher uses a different substitution for each letter in the plaintext.
 - Repeating eventually.
 - A mechanical encryption device can implement this easily with rotors (Rotor Machine).

6. Running Key ciphers:

- A running key cipher takes this to the extreme with a key just as long as the plaintext.
- A short key generator is often used such as a pseudo-random number generator.



Lecture 3:

Overview



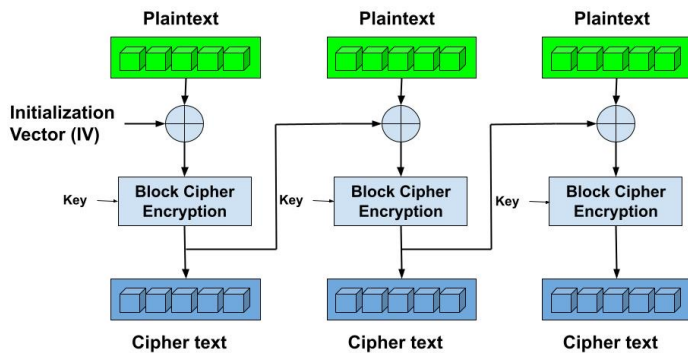
Outline

1. Feistel ciphers
 - A family of single-key, block substitution ciphers
2. Data Encryption Standard (DES)
 - 64-bit Block cipher, 56-bit key length
 - Insecure for modern applications
 - Still widely used
3. Advanced Encryption Standard (AES)
 - 128-bit Block cipher, different key lengths (128, 192, 256)
 - Replacement for DES



Block Ciphers

Block Cipher



Key Parameters	Stream Cipher	Block Cipher
Definition	A stream cipher, on the other hand, breaks down a plaintext message into single bits, then encrypts one byte of the message at a time.	A block cipher is an encryption technique that breaks down plaintext messages into fixed-size blocks then converts one block of the message at a time.
Conversion of Bits	Stream cipher uses 8 bits.	In other hand, Block cipher uses either 64 bits or more than 64 bits.
Speed	Stream cipher is fast in comparison to block cipher.	Block cipher is slow as compared to a stream cipher.
Techniques Used	Stream Cipher uses substitution techniques.	While Block Cipher uses transposition techniques.
Key Used	One key is used only one time in stream cipher.	One key is used multiple times in block cipher.
Confusion and Diffusion	It Relies only confusion .	It Uses both confusion and diffusion.
Implementation	The main implementation of Stream Cipher is Vernam Cipher.	While on the other side, The main implementation of Block Cipher is Feistel Cipher.
Reversibility	Due to use of XOR for the encryption, reverse encrypted text is easy.	In block cipher, reverse encrypted text is hard.

<https://cybermeteoroid.com/stream-cipher-and-block-cipher-a-complete-overview/>



Feistel Ciphers

- This is a family of single-key, block substitution ciphers.
- Feistel was in charge of the IBM Lucifer project (1973) and provided the theoretical underpinning for many of the first block ciphers.
 - His work was based on Shannon (1945)
- He proposed 64-bit or 128-bit block sizes.



Feistel Ciphers: the Keys

- The key is the mapping between input and output blocks.
- If all possible mappings are possible then the keys would be 10^{89} bits long for a block size of 64.
 - Stirling's approximation of $64!$.
 - There are thus a lot of possible keys.
- The actual size of the key space was reduced, with a key length of 128 bits.
 - Still 2^{128} or approximately 10^{38} different keys.



Feistel Ciphers: Product Ciphers

- Feistel used a *product cipher*.
 - Several small transformations are applied one after the other.
- He alternated substitutions and transpositions.
- The resulting large transformation is likely to be much harder to break than each of the individual transformations.
- Each substitution uses a sub-key, which is generated from the master key.



Feistel Ciphers: the Structure

- The initial data block is split into two halves, the left (L_0) and right (R_0) halves.
 - Actual operations involve just one-half of each data block and so fewer bits.
- Computer word size at the time was 32 bits.
 - Each half block was 32 bits and so fitted in one computer word.
 - This sped up data processing.
- There are a number of rounds n , each making a substitution followed by swapping the two halves (the transposition).



Feistel Ciphers: Encryption Algorithm

Encrypt plaintext (L_0, R_0) :

- For each round $i = 0, 1, 2, \dots, n$:
 - Input: L_i , R_i , and K_i (round key)
 - Output: L_i and R_i
 - A substitution is performed on the left half L_i

$$L_i \oplus F(R_i, K_i)$$

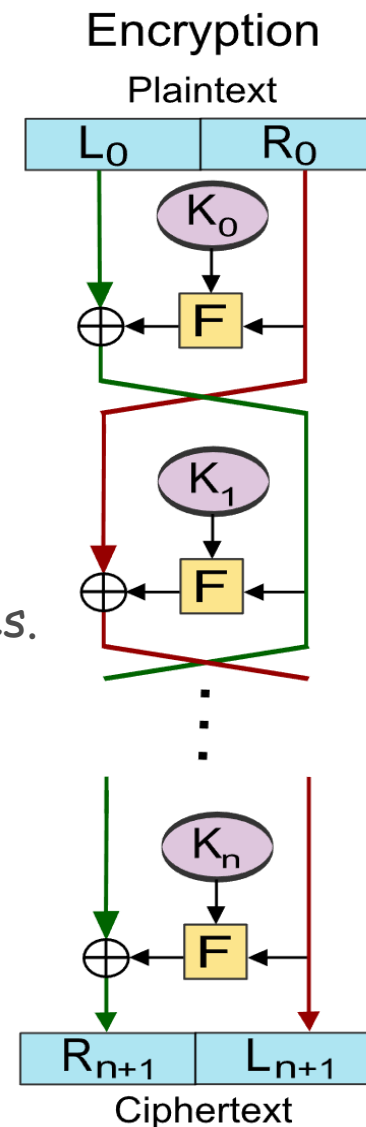
- A permutation is performed by swapping the two halves.

$$L_{i+1} = R_i$$

$$R_{i+1} = L_i \oplus F(R_i, K_i)$$

- Then the ciphertext is (R_{n+1}, L_{n+1})

- ❖ F is any function, there are several different Feistel algorithms, each with a different function F .
- ❖ K_i is a round key (i.e., sub-key), $\oplus$ is Exclusive OR.





Feistel Ciphers: Design Features

- **Block size:** The larger it is, the more secure the cipher is but the slower the cipher is.
 - 64 or 128-bit
- **Key size:** The larger it is, the more secure the cipher is but the slower the cipher is.
 - 64 or 128 bits
- **Number of rounds n :** The larger it is, the secure the cipher is but the slower the cipher is.
 - 16 rounds is typical.
- **Sub-key Generation K_i :**
 - The more complex it is, the more secure the cipher is but the slower the cipher is.
- **Function F :**
 - The more complex it is, the more secure the cipher is but the slower the cipher is.



Feistel Ciphers: Decryption Algorithm

Decrypt ciphertext (R_{n+1}, L_{n+1}) :

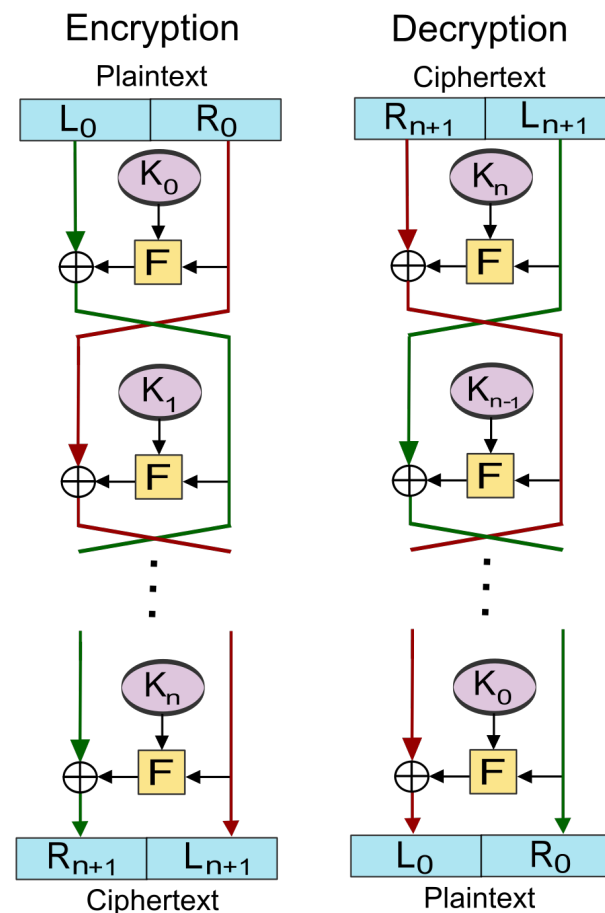
- For each round $i = n, n-1, \dots, 0$:

$$R_i = L_{i+1}$$

$$L_i = R_{i+1} \oplus F(L_{i+1}, K_i)$$

Then the plaintext is (L_0, R_0) again

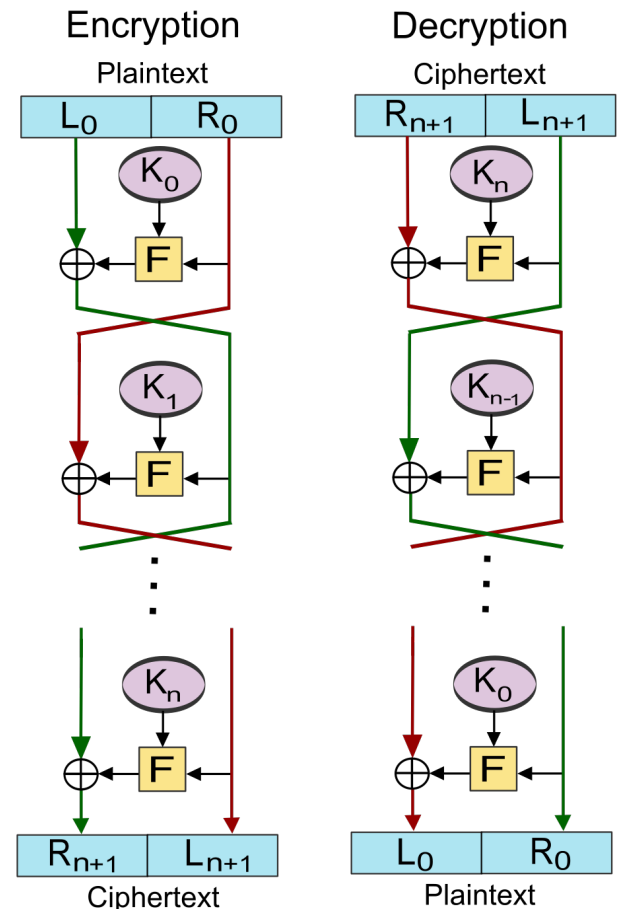
- The diagram illustrates both Encryption and Decryption processes
- Note the reversal of the subkey order for decryption: this is the only difference between encryption and decryption.





Feistel Ciphers: Decryption Algorithm

- Decryption follows the same steps but with the sub-keys used in the **reverse order**.
- The same algorithm **F** and keys are used for encryption and decryption
- To prove this works, all we need to do is show that two applications of each substitution step, with the same key, cancel each other.
- From the formula: Swapping left and right halves will cancel each other out.
 - Swap two items and then swap them again. They are back in the same place.





Feistel Ciphers: Proof

We're going to prove that Two substitutions cancelled
Consider any round i :

- Let $B_i = (L_i, R_i)$ be a block, split into left and right halves.
 $B_i = (L_i, R_i)$ is the initial plaintext block.
 $B_{\text{encrypt}} = (L_{\text{encrypt}}, R_{\text{encrypt}})$ is ciphertext after encryption.
 $B_{\text{decrypt}} = (L_{\text{decrypt}}, R_{\text{decrypt}})$ is plaintext after decryption.
- All we need do is show that $L_{\text{decrypt}} = L_i$ and $R_{\text{decrypt}} = R_i$



Feistel Ciphers: Proof

We're going to prove that two substitutions cancelled

➤ Applying the Feistel function, we have:

- For Encryption:

$$L_{\text{encrypt}} = R_i$$

$$R_{\text{encrypt}} = L_i \oplus F(R_i, K_i);$$

The ciphertext to be decrypted is $(L_{\text{encrypt}}, R_{\text{encrypt}})$

- For Decryption

$$R_{\text{decrypt}} = L_{\text{encrypt}} = R_i$$

$$\begin{aligned} L_{\text{decrypt}} &= R_{\text{encrypt}} \oplus F(L_{\text{encrypt}}, K_i) \\ &= L_i \oplus F(R_i, K_i) \oplus F(L_{\text{encrypt}}, K_i) \\ &= L_i \oplus F(R_i, K_i) \oplus F(R_i, K_i) \\ &= L_i \oplus 0 = L_i \end{aligned}$$



Feistel Ciphers: Function F Design Criteria

- No output bit is close to a linear function of a subset of the input bits.
- If they were it would be vulnerable to a chosen plaintext attack (linear cryptanalysis).
 - Choose several plaintext messages whose bits form a linear combination
 - Solve the linear equations.
- There is no bias towards some bit positions.
- If two different inputs (key or data) differ by 1 bit, the output must differ by at least two bits.
 - Confusion and diffusion.



Summary

- The Feistel family of algorithms were designed to be fast.
 - The block size was two computer words. All the calculations were done in a single computer word of 32 bits (half of the block).
 - There were several rounds with sub-keys generated from the master key.
 - In each round:
 - A substitution is performed on the left half L_i
 - A permutation is performed by swapping the two halves.
 - Decryption used the same algorithm but with the subkeys in the reverse order.